

1. Let $A = \{a, b, c, d\}$ and $B = \{a, b, e, f, g\}$
 - a. Find $A \cap B$
 - b. Find $A \cup B$.
 - c. Find $B - A$.
 - d. Find $A - B$.

2. Let A be a nonempty subset of B .
 - a. Prove that $A \cup B = B$.
 - b. Prove that generally $A \cap B = B$ is false.

3. Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$.
 - a. Find all possible subsets of A .
 - b. List 2 elements in $A \times B$.

4. Prove that $A \cup \emptyset = A$.

5. Prove that $A \cap \emptyset = \emptyset$.

6. Consider the map $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = 2x + 1$. Prove that f is onto.

7. Define the relation $\mathcal{R}$ on the real number set $\mathbb{R}$ as $x\mathcal{R}y \iff x \geq y$.
 - a. Show that $\mathcal{R}$ is reflexive
 - b. Show that $\mathcal{R}$ is symmetric.
 - c. Show that $\mathcal{R}$ is transitive.
 - d. Is $\mathcal{R}$ an equivalence relation?

8. Give an example of a relation which is not transitive.

9. Let X be a nonempty set. Define the following relation on the set of subsets of X as follows $A\mathcal{R}B \iff A \cap B \neq \emptyset$ for $A, B \subseteq X$. Prove that $\mathcal{R}$ is not reflexive.

10. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions.
 - a. Suppose that $|A| > |B|$. Prove that f is not one-to-one.
 - b. Suppose that $|A| < |B|$. Is f onto?
 - c. Is $g \circ f$ onto then is g onto? is f onto?

11. Let S_3 be the symmetric group on three elements.
- Is S_3 abelian?
 - Is S_3 cyclic?
 - Prove that S_3 has an abelian subgroup.
 - Prove that S_3 has a non-abelian subgroup.
 - What is the order of S_n .
12. Give the definition of homomorphism, and show by example that $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = 2x$ is a group homomorphism (under addition).
13. Let $f: G_1 \rightarrow G_2$ be a group homomorphism and let $H \leq G_2$. Recall that $f^{-1}(H) = \{g \in G_1 \mid f(g) \in H\}$. Prove that $f^{-1}(H) \leq G_1$.
14. Let G be a finite group of order 24 and $g \in G$.
- List the *possible* orders of g .
 - Let H be a subset of G such that $|H| = 7$. Is $H \leq G$?
 - Let $|H| = 8$. Does Lagrange's theorem imply that $H \leq G$?
15. Let G be a group of order 4. Prove that G is abelian.
16. Let G be a group where 0_G is the identity.
- Find a subgroup of G .
 - Find a subgroup of G distinct from the group given for part (a).
 - If $|G| = 10$ and $H \leq G$. Prove that $|H| \neq 6$
17. Let H and K be subgroups be non-trivial subgroups of G . Prove that $H \cap K$ is a subgroup of G .
18. Let R be a ring and $a, b \in R$.
- Which statement is true?
 - (a) If $ab = 0$ then $a = 0$ or $b = 0$.
 - (b) If $a^2 = 0$ then $a = 0$.
 - (c) If $a \neq 0$ and $b \neq 0$ then $ab \neq 0$.
 - (d) None of the above.
19. Prove that $\mathbb{Z}$, the set of integers, is a ring under addition and multiplication.